# Sahaj Bhadja

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## Summary

Cloud software engineer with 2+ years building secure SaaS services, distributed workflows, and production APIs across AWS and Azure. Skilled in Python, Java, microservices, CI/CD, observability, and reliability engineering.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Architected multi-tenant **Azure SQL** and Blob Storage services for 500+ users, applying row-level security, indexed queries, and secure access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **distributed workflows** for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading service failures.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Engineered Python Flask services and **REST APIs** handling 10,000+ daily requests, integrating secure file ingestion, AWS RDS persistence, and concurrent processing while maintaining stable production performance during peak workloads.
- Implemented multithreaded batch workflows for 1,000+ files with validation, error isolation, structured logging, and **CI/CD** releases, reducing feedback turnaround by 60% while improving operational reliability for 150+ instructors.

**USC AD Labs** **Los Angeles, CA**
*Machine Learning Researcher* *January 2025 – May 2025*
- Built Python research pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that improved debugging, benchmarking, and experiment traceability for algorithmic system analysis.
- Documented experimental behavior, failure cases, and reporting outputs for researchers, translating complex service signals into actionable findings while maintaining clear validation procedures and reproducible technical artifacts for team review.

**Aasma Technology Solutions** **India**
*Software Developer Engineer – SecureSync* *August 2023 – July 2024*
- Engineered cross-platform **microservices** with Jenkins pipelines, automated VM builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening release quality and production service readiness.
- Optimized large-scale API and ETL workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving response times by 30% and production pipeline efficiency by 68%.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern – VibrantChem* *May 2023 – August 2023*
- Developed REST API and WebSocket workflows synchronizing 2M+ records across modular services, improving consistency and real-time visibility through validation controls, fault handling, and responsive web client interfaces.
- Automated Docker-based build and deployment pipelines with Jenkins CI/CD, unit validation, and release checks, reducing deployment time by 50% while improving repeatability across development and production environments.

## Skills

**Languages:** Python, Java, JavaScript, TypeScript, SQL, C++, Bash, C#
**Cloud & Platform:** AWS, Azure, Google Cloud, Docker, Kubernetes, microservices, SaaS, distributed systems
**Backend & Data:** FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB
**DevOps & Reliability:** Jenkins, GitHub Actions, CI/CD, Git, observability, metrics, logs, alerts, root-cause analysis
**Security & Quality:** RBAC, identity and access control, secure APIs, secrets management, PyTest, Mockito, unit testing, integration testing, Agile

## Projects

**FinNexus** | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a containerized fintech platform supporting 50,000+ users and 1M+ transactions with token-based access controls, real-time synchronization, resilient APIs, service monitoring, and Kubernetes-backed CI/CD deployment automation.

**Linguistix** | *Python, AWS, Docker, CodePipeline, REST APIs*
- Developed containerized language-processing services with AWS CodePipeline releases, automated validation, structured error handling, and scalable REST interfaces, achieving 87% accuracy while improving repeatable integration across production environments.

**Blockchain Authentication** | *Python, JavaScript, IoT, RBAC*
- Implemented decentralized authentication and four-tier role-based access control for secure IoT workflows, reducing repeated key-verification operations by 50% while strengthening identity enforcement, service authorization, and tamper-resistant auditability.